

Bony alignment decisions affect patient-specific laxity phenotype patterns significantly, independent of the deformity

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Abstract

Purpose: While bony alignment phenotype reconstruction became an important part of personalised knee arthroplasty, the knowledge on laxity phenotypes (LPs) is still limited. This study aimed to calculate individual LPs and assess their changes based on bony decisions from different alignment workflows.

Methods: Radiographs and computer-assisted surgery data of 86 knees were imported into a validated knee alignment simulator. Individual bony parameters (medial proximal tibial angle, lateral distal femoral angle and posterior condyle axis) were first introduced. By that, the patient-specific bony phenotype (B-FKP) was implemented, and based on these simulations, the patient-specific laxity phenotype (L-FKP) was defined, calculated and analysed for the total group, as well as for all Coronal Plane Alignment of the Knee (CPAK) subgroups. CPAK I and IV were summarised as varus group; II and V as neutral, and III, VI and IX as valgus group. Identical calculations were then compared for the MA and L-FKP of both workflows. LP was calculated in both extension (L-FKPext) and flexion (L-FKPflex), and a pattern matrix was constructed for all possible L-FKP combinations, enabling a comprehensive distribution analysis. Statistical differences between subgroups and B-FKP and mechanical alignment (MA) workflows were calculated.

Results: B-FKP showed a minimal, non-significant difference for L-FKPext in all subgroups; however, a huge variability in L-FKP pattern analysis. In contrast, MA showed a significant difference for L-FKPext in all subgroups, with a high correlation between L-FKPext and hip–knee–ankle angle. While in MA, 98% of knees showed lateral laxity (L-FKPflex-lat), in B-FKP, only 56% were L-FKPflex-lat, with a large variability (31% L-FKPflex-neutr and 13% L-FKPflex-med).

Abbreviations: B-FKP, bony-functional knee phenotypes; CPAK, Coronal Plane Alignment of the Knee; LDFA, lateral distal femoral angle; LP, laxity patterns; L-FKP, laxity-functional knee phenotypes; L-FKPext, laxity-functional knee phenotype in extension; L-FKPext-neutr, neutral laxity-functional knee phenotype in extension; L-FKPext-valg, valgus laxity-functional knee phenotype in extension; L-FKPext-var, varus laxity-functional knee phenotype in extension; L-FKPflex, laxity-functional knee phenotype in flexion; L-FKPflex-lat, lateral laxity in flexion laxity-functional knee phenotype; L-FKPflex-med, medial laxity in flexion laxity-functional knee phenotype; L-FKPflex-neutr, neutral laxity in flexion laxity-functional knee phenotype; MA, mechanical alignment; MPTA, medial proximal tibial angle; PCA, posterior condylar axis.

Conclusions: Personalised bony resections reduce gap differences for LPext and LPflex independent of the deformity. MA showed a high correlation between deformity and LPext in extension and a uniform lateral laxity in flexion. L-FKP analysis can help to understand the individuality of knees from a soft tissue aspect.

Level of Evidence: Level III.

KEYWORDS

alignment, bony phenotype, CPAK, knee, laxity phenotype, mechanical alignment

INTRODUCTION

In the last few decades, personalised alignment techniques have gained preference in total knee arthroplasty (TKA) due to functional patient dissatisfaction associated with mechanical alignment (MA). Studies have shown that nearly 20% of patients report suboptimal functional outcomes, likely due to inadequate restoration of native knee kinematics [1, 8]. Altering individual bony anatomy was identified to be one of the main reasons for limited patient satisfaction and reduced function. To achieve a better individual bony reconstruction, a pre-operative analysis of various joint parameters such as medial proximal tibia angle, joint line obliquity or distal femoral angle became a new pre-operative standard [16]. The two classification systems that are currently mostly used, CPAK [7, 18, 30] and functional knee phenotypes [2, 9–12], allow for a more detailed analysis of these parameters.

Traditional MA techniques aim to achieve equal flexion and extension gaps medially and laterally [14]. However, multiple studies [3, 5, 6] indicate that native knee biomechanics rarely conform to such uniform gap balancing. This points out that not only the bony functional knee phenotype (B-FKP) of a knee but also the laxity phenotype (L-FKP) might be altered by MA in a considerable number of patients. Due to this observation, some alignment workflows accept or even aim to achieve a larger lateral flexion gap [15, 20]. This, however, is again a systematic approach, making all knees the same, ending with a larger lateral flexion gap. This is probably not identical with a reconstruction of the patient-specific L-FKP. To better understand the L-FKP of a knee, a classification similar to the B-FKP classification would be helpful. So far, no such classification system has been reported, and therefore, it is still unclear which knee needs which balancing target.

The patient-specific knee laxity needs to be defined using a laxity classification in a standardised manner to be able to calculate it in every knee. It is clear that this L-FKP calculation must include the L-FKP in extension as well as in flexion, as both might be different in many knees [3, 5, 6, 17]. Another important aspect for L-FKP

calculation is the patient-specific B-FKP, as B-FKP determines joint line obliquity and by that directly affects the tension of the collateral ligaments.

The aim of this study was to (1) define and calculate L-FKP patterns in extension (L-FKPext) and flexion (L-FKPflex) and to (2) analyse the effect of bony resections of different alignment workflows on L-FKP patterns. This was performed by simulating bone cuts in (a) patient-specific B-FKP (medial proximal tibial angle [MPTA] and lateral distal femoral angle [LDFA]) and in (b) the MA workflow and calculating the following laxity patterns. The influence of deformity on L-FKP calculation was analysed by comparing varus, valgus and straight legs within different CPAK groups.

The following specific hypothesis was formulated:

Bony resections affect individual patient-specific laxity patterns (L-FKP) in both, extension and flexion significantly, independent of the deformity type.

MATERIALS AND METHODS

Definition and calculation of L-FKP

For calculating L-FKPext and L-FKPflex, the Knee-CAT simulator (SOS GmbH) was used, and data from 86 consecutive, previously operated knees were analysed. Inclusion criteria were patients with OA IV°; no previous ligament surgeries or known traumatic events, no FFD > 15°, no deformity > 20°, knee flexion of 90° or more, and no previous infection. No patient had previous surgery with the exception of knee arthroscopy for degenerative meniscal tears. All data sets were obtained during previous computer-assisted surgery (CAS) by the first author. Therefore, the present study is based on simulating laxity in extension and flexion (90°) based on previously recorded gap measurements. All data is derived from a retrospective, single cohort for which all patients have signed and approved the General Consent for retrospective use of operative data. The present study follows the principles outlined by the Declaration of Helsinki [31], and each step performed during the study (data collection, analysis, data reporting, etc.) was performed by adhering to

Good Clinical Practice guidelines [4]. Patient-specific B-FKP was calculated from pre-operative long leg standing X-rays with digital measurement of MPTA and LDFA according to Paley et al. [24]. Gap data and data on the bony surface model were obtained during CAS (Knee 3.0; Brainlab). Standardised gap analysis was applied in extension and flexion according to the formerly validated stress testing method [21]. Gap analysis was performed after removal of osteophytes to reduce the potential, pathological tensioning on the tested ligaments. As this intraoperative measurement was performed before any bone cuts were performed, removal of posterior tibial and femoral osteophytes was not always completely done. All simulations were performed using a symmetrical femoral implant with an implant thickness of 9 mm distally and posteriorly (Attune PS, DePuy Medtech).

During TKA surgery, gap measurements were initially displayed using the MA default setting (Knee 3.0; Brainlab). The projected values were derived from standardised cutting parameters, including 3° external rotation relative to the posterior condyles, while the distal femoral cut and proximal tibial cut were both 0° relative to the mechanical axis. Default setting for posterior tibia slope was 5°. These parameters directly influenced gap dimensions and, consequently, L-FKP in both, extension and flexion (Figure 1a). For analysis

of individual L-FKP, the values of the patient-specific B-FKP (MPTA, LDFA) were implemented first in the simulation. Boundaries were implemented for MPTA (85°–92°), LDFA (86°–92°) and hip–knee–ankle angle (HKA) (175°–183°). To stay within HKA boundaries, sometimes tibia and/or femoral cuts need to be restricted below their defined boundaries. In extension, both cuts were simulated with the individual proximal tibia cut (MPTA) as well as the individual distal femoral cut (LDFA) (Figure 1b). The altered gap values were recorded.

Therefore, L-FKP in extension (L-FKP_{ext}) is defined as the difference between medial and lateral extension gap after incorporating patient-specific B-FKP (MPTA and LDFA). In flexion, only MPTA, but not LDFA affects gap values. The other bony reference for calculating the individual flexion gap was the posterior condyle axis (PCA). Both posterior condyles should be cut equally to restore the individual bony anatomy in a symmetric implant. The cartilage loss in osteoarthritic knees was included based on intraoperative thickness measurements (calliper technique), and a standard thickness of 2 mm was considered in the unworn compartments. Therefore, L-FKP in flexion (L-FKP_{flex}) is defined as the difference between medial and lateral flexion gap after dialling in MPTA and rotating the femur until parallel posterior cuts are achieved.

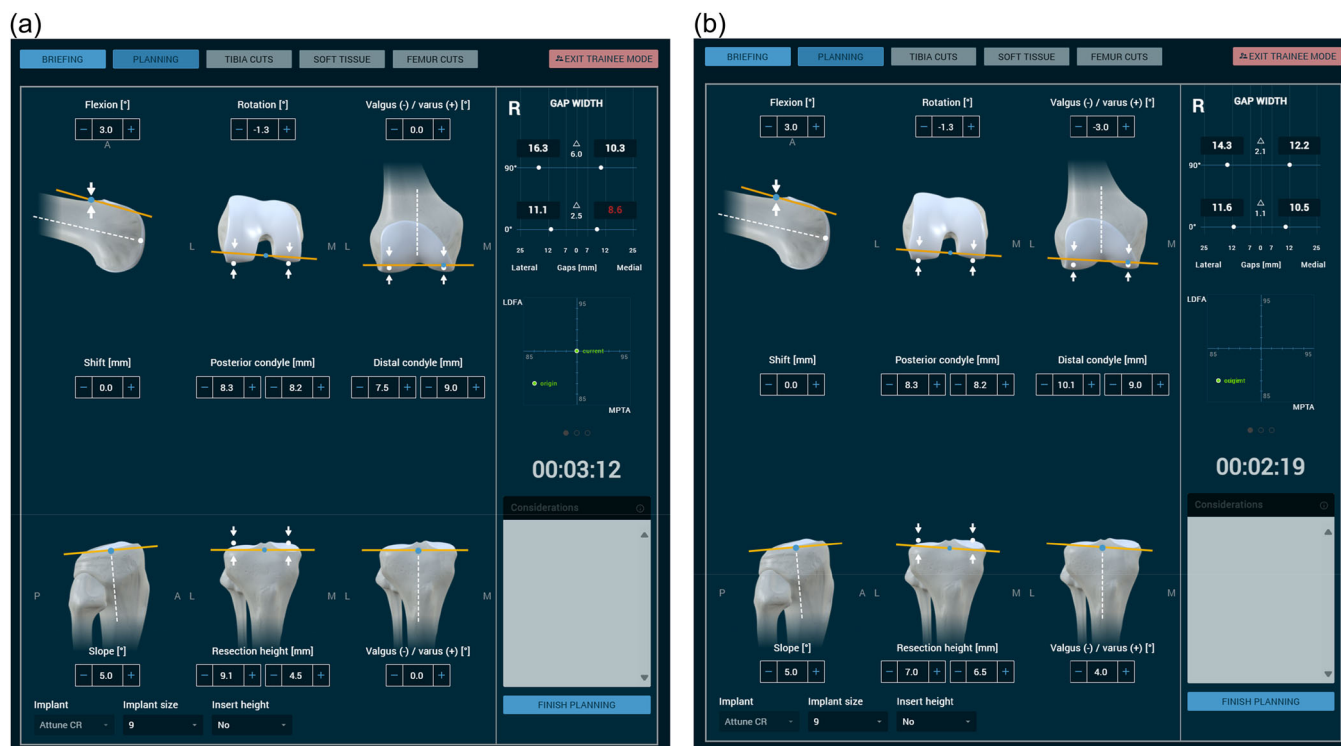


FIGURE 1 (a) Default setting for planning according to MA definitions. Femoral rotation corrected to equal posterior condyle cut. (b) Adapted planning screen after implementing patient-specific B-FKP (MPTA, LDFA). Femoral rotation corrected to equal posterior condyle cut. Based on these adaptations, the patient-specific L-FKP is displayed. B-FKP, bony-functional knee phenotypes; L-FKP, laxity-functional knee phenotypes; LDFA, lateral distal femoral angle; MA, mechanical alignment; MPTA, medial proximal tibial angle.

L-FKPext is defined as neutral (L-FKPext-neutr) if the medial-lateral gap difference is between -1 and $+1$ mm; L-FKPext is varus (L-FKPext-var), if the lateral gap is more than 1 mm larger than the medial extension gap. L-FKPext is valgus (L-FKPext-valg) if the lateral extension gap is more than 1 mm smaller than the medial extension gap (Figure 2a).

L-FKPflex is defined as neutral (L-FKPflex-neutr) if the medial-lateral flexion gap difference is between -1 and 1 mm. L-FKPflex is showing lateral laxity (L-FKPflex-latlax)

if the lateral flexion gap is more than 1 mm larger than the medial flexion gap. L-FKPflex is showing medial laxity (L-FKPflex-medlax) if the medial flexion gap is more than 1 mm larger than the lateral flexion gap (Figure 2b).

To analyse laxity patterns of all knees, a matrix of all options for L-FKPext (varus, straight and valgus) and L-FKPflex (lateral laxity; neutral, medial laxity) was created, based on the definitions mentioned above. Nine potential patterns exist, and all knees were grouped according to their L-FKPs.

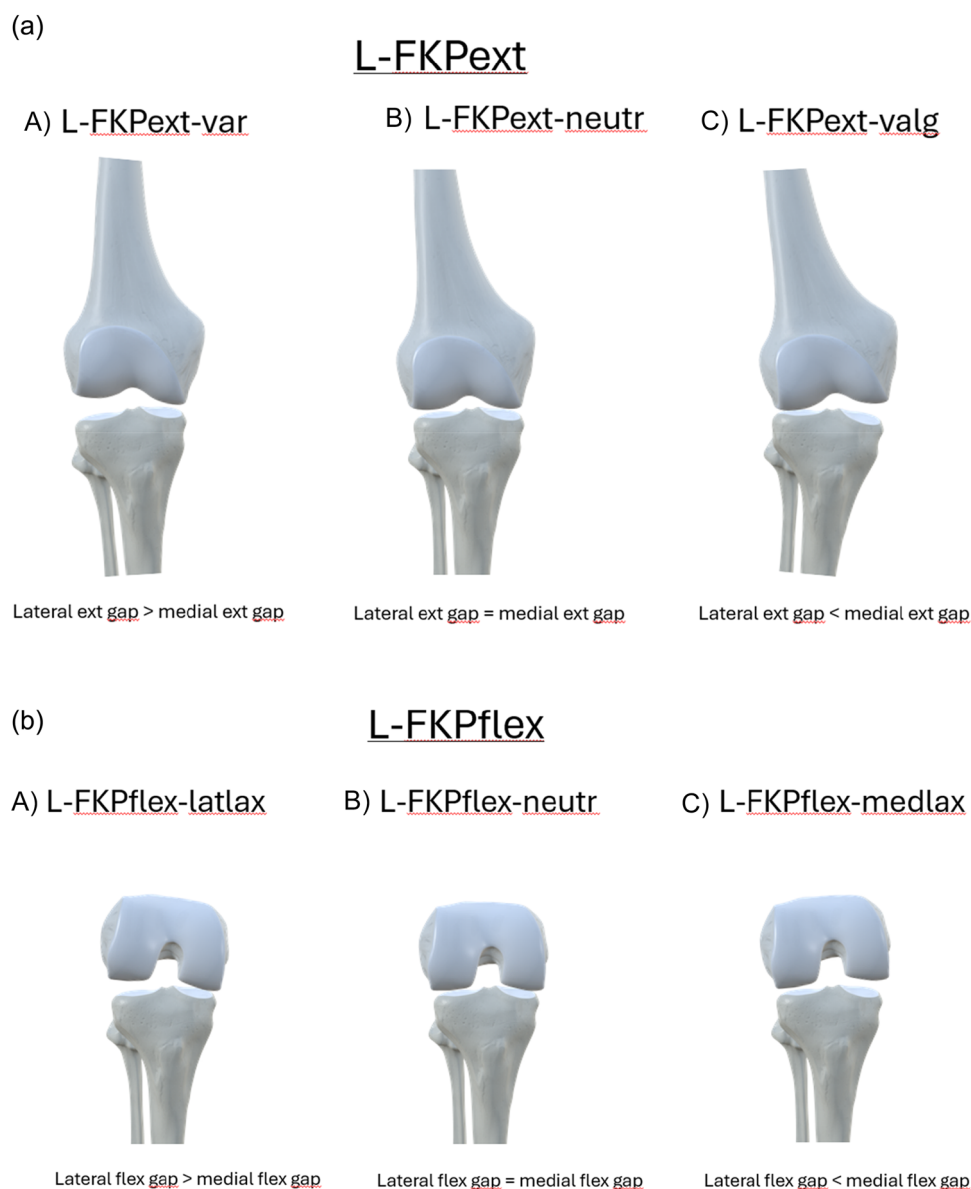


FIGURE 2 (a) Showing the three different L-FKP in extension. (A) L-FKPext-var is present if the lateral extension gap is more than 1 mm larger than the medial extension gap. (B) L-FKPext-neutr is present if medial and lateral extension gaps are equal ± 1 mm. (C) L-FKPext-valg is present if the lateral extension gap is more than 1 mm smaller than the medial extension gap. (b) Showing the three different L-FKP in flexion. (A) L-FKPflex-latlax is present if the lateral flexion gap is more than 1 mm larger than the medial flexion gap. (B) L-FKPflex-neutr is present if medial and lateral flexion gaps are equal ± 1 mm. (C) L-FKPflex-medlax is present if the lateral flexion gap is more than 1 mm smaller than the medial flexion gap. L-FKP, laxity-functional knee phenotypes; L-FKPext, laxity-functional knee phenotype in extension; L-FKPflex, laxity-functional knee phenotype in flexion; L-FKPflex-neutr, neutral laxity in flexion laxity-functional knee phenotype; L-FKPext-valg, valgus laxity-functional knee phenotype in extension; L-FKPext-var, varus laxity-functional knee phenotype in extension.

Analysis of L-FKP changes between B-FKP workflow and MA

For calculating individual L-FKPext and L-FKPflex, the patient-specific B-FKP was set on the simulation screen. This was defined as B-FKP workflow. In a first step (a), based on these calculations, the patient-specific L-FKP was recorded according to the aforementioned definitions. This was done for the entire group of simulated knees as well as for each of the CPAK groups. Based on CPAK, knees were classified as varus knees if they belonged to CPAK I and IV, straight legs (CPAK II and V) or valgus knees (CPAK III, VI and IX). Distribution of L-FKP was calculated for L-FKPext and L-FKPflex according to the above-mentioned definitions. Mean values and standard deviations were calculated for each of these groups. Correlation between deformity (HKA) and L-FKP (L-FKPext and L-FKPflex) was performed for each B-FKP.

For assessing the effect of altered bony cuts on L-FKP, in a second part (b) MA was simulated by setting a standardised 0° cut for the proximal tibia and the distal femur in all knees as well as by implementing all standard settings (eg. 3° femoral flexion, 5° tibial slope). Results of L-FKP with B-FKP and MA workflows were compared for extension (L-FKPext) and flexion (L-FKPflex). Distribution of L-FKP was calculated and compared with B-FKP workflow. Correlation between deformity (HKA) and L-FKP in MA was performed for L-FKPext and L-FKPflex.

Statistics

All data were analysed using Microsoft Excel. Descriptive statistics, including means and standard deviations, were calculated for the following variables: MPTA, LDFA, HKA and gap sizes for L-FKPext and L-FKPflex in B-FKP and MA.

To assess the relationships between variables, Pearson correlation coefficients were calculated for the following pairs: HKA and L-FKPext as well as HKA and L-FKPflex in B-FKP and MA, HKA and gap sizes in MA, and HKA and gap sizes in B-FKP. Correlations were calculated for the entire cohort and separately for the Varus, Valgus, and Straight subgroups.

To determine if there were significant differences between L-FKPext in MA and B-FKP, and between gap sizes in MA and B-FKP, regression analyses were performed using Excel's regression tool. These analyses were conducted for the entire cohort and separately for the Varus, Valgus, and Straight subgroups.

Independent samples *t* tests assuming equal variances were used to compare the means of flexion and extension angles between Varus, Valgus, and Straight subgroups, as well as between each subgroup and the overall group. For these comparisons, MA values were compared against MA values, and individual alignment

values were compared against individual alignment values.

Post hoc power analysis assessed the strength of the observed Pearson correlations (*r*) based on sample size, effect size (*r*), and a significance level of 0.05. Effect sizes (Cohen's *d*) were also calculated for all independent samples *t* tests using mean $\pm$ SD.

RESULTS

Knee characteristics

Eighty-six knees were analysed in total. Subgroups were analysed according to definitions in CPAK groups. Forty-one knees were rated straight (27 CPAK II and 14 CPAK V), 32 knees as varus (26 CPAK I and 6 CPAK IV) and 13 valgus (7 CPAK III/5 CPAK VI and 1 CPAK IX). Twenty-five knees showed a horizontal joint line, 50 an oblique joint line apex distal and only 1 knee with apex proximal. Bony characteristics for HKA, MPTA and LDFA are summarised in Table 1.

LP patterns and calculations

LP patterns and calculation after patient-specific bony phenotype reconstruction (B-FKP workflow)

Huge variations for L-FKP patterns were found (Table 2). After simulation of patient-specific B-FKP reconstruction, 44% of cases were already balanced (I-FKPext-straight) in extension without any release. Due to boundaries in CPAK I, almost 40% were still left in L-FKPext-var. Similar was found in most of CPAK III for valgus knees. The majority of knees (56%) showed a lateral laxity (L-FKPflex-lat), while 31% were neutral (L-FKPflex-neut) and 13% showed a medial laxity (L-FKPflex-med).

A low correlation ($r=0.28$) between HKA and L-FKPext suggests that coronal knee alignment alone is insufficient to predict laxity patterns, with a resulting

TABLE 1 Showing the bony characteristics (MPTA, LDFA) and leg axis (HKA) for the total group, as well as for varus, straight and valgus knees.

	HKA	MPTA	LDFA
Total (<i>n</i> = 86)	175.2 $\pm$ 5.9°	87.0 $\pm$ 3.0°	88.1 $\pm$ 2.3°
Varus (<i>n</i> = 32)	168.5 $\pm$ 7.8°	84.5 $\pm$ 1.8°	89.7 $\pm$ 1.5°
Straight (<i>n</i> = 41)	176.9 $\pm$ 3.9	87.6 $\pm$ 1.9°	87.9 $\pm$ 1.9°
Valgus (<i>n</i> = 13)	183.1 $\pm$ 4.1	90.4 $\pm$ 2.3	85.0 $\pm$ 2.0

Abbreviations: LDFA, lateral distal femoral angle; HKA, hip–knee–ankle angle; MPTA, medial proximal tibial angle.

statistical power of 0.85. Similarly, the negative correlation ($r = -0.14$) between HKA and L-FKPflex indicates that additional factors play a role in knee laxity. Nonetheless, the resulting statistical power was 0.36, suggesting that the sample size was not adequate to accurately investigate the correlation between HKA and flexion laxity. No significant differences for L-FKPext were found between subgroups, showing an almost balanced knee in all groups after placing the implant cuts according to the patient-specific bony characteristics (Table 3a).

L-FKPflex showed in all groups a larger lateral flexion gap (Table 3a). No significant differences were found between varus and straight ($d = 0.12$) as well as between varus and valgus ($d = 0.76$). The only significant difference was found between straight and valgus ($p < 0.05$, $d = 0.68$), the difference being larger for valgus.

A negative, statistically insignificant, correlation between L-FKPext and L-FKPflex was found ($r = -0.14$) with a resulting post-hoc statistical power of 0.25.

TABLE 2 Laxity patterns for extension gap (varus/straight/valgus) and flexion gap (lateral laxity/neutral and medial laxity) combinations.

I	Lateral laxity	Neutral	Medial laxity	Total
Varus	22	6	2	30
Straight	20	12	6	38
Valgus	6	9	3	18
Total	48	27	11	86

Note: The Biggest groups are varus in combination with lateral laxity and straight in combination with lateral laxity.

TABLE 3a L-FKPext and L-FKPflex calculations for all groups with simulation of bone cuts based on patient-specific bony phenotypes (B-FKP).

	L-FKPext (B-FKP)	L-FKPflex (B-FKP)	Fem. Rotation
Total	$0.4 \pm 2.0^\circ$	$1.8 \pm 2.8^\circ$	$-0.3 \pm 0.8^\circ$
Varus	$0.2 \pm 2.1^\circ$, n.s. 11 (var)-12 (straight)-8 (valg)	$2.1 \pm 1.6^\circ$, n.s. 6 (ir)-10 (neut)-16 (er)	$-0.5 \pm 0.6^\circ$
Straight	$0.5 \pm 2.1^\circ$, n.s. (14-20-4)	$1.8 \pm 3.0^{**}$ (5-14-22)	$-0.3 \pm 1.0^\circ$
Valgus	$0.1 \pm 1.9^\circ$, n.s. (5-5-3)	$3.7 \pm 2.5^{**}$ (0-4-9)	$0.1 \pm 0.7^\circ$

Note: No significant differences between subgroups for L-FKPext. L-FKP flex only significant differences (*) between valgus and straight, but not between the others. Huge variations of L-FKP patterns within groups, for L-FKPext as for L-FKPflex.

Abbreviations: B-FKP, bony-functional knee phenotypes; L-FKPext, laxity-functional knee phenotype in extension; L-FKPflex, laxity-functional knee phenotype in flexion.

TABLE 3b L-FKPext and L-FKPflex calculations for all groups by simulation bony cuts based on MA.

	L-FKPext (MA)	L-FKPflex (MA)	Fem. Rotation
Total	$1.6 \pm 3.2^\circ$ (47-22-17)	$4.5 \pm 2.8^\circ$ (1-1-84)	$-0.3 \pm 0.8^\circ$
Varus	$4.2 \pm 2.4^{**}$ (31-1-0)	$5.6 \pm 2.4^{**}$ (0-1-31)	$-0.5 \pm 0.6^\circ$
Straight	$1.0 \pm 2.0^{**}$ (16-19 -6)	$4.1 \pm 3.0^{**}$ (1-0-40)	$-0.3 \pm 1.0^\circ$
Valgus	$-2.9 \pm 1.8^{**}$ (0-2-11)	$3.6 \pm 1.6^\circ$ (0-0-13)	$0.1 \pm 0.7^\circ$

Note: Significant differences (*) between all subgroups for L-FKPext, and for L-FKPflex, except for L-FKPflex between straight and valgus knees. While L-FKPext showed a change of varus/valgus/straight distribution within groups according to the deformity (HKA), there were no differences between groups for L-FKPflex.

Abbreviations: HKA, hip-knee-ankle angle; L-FKPext, laxity-functional knee phenotype in extension; L-FKPflex, laxity-functional knee phenotype in flexion; MA, mechanical alignment.

LP calculation after MA

HKA correlated highly with the L-FKPext in MA ($r = 0.83$) with a post hoc statistical power of >0.85 (Table 3b). L-FKPext in the varus group was significantly larger than in the straight and valgus groups, while for straight knees it was significantly larger than in the valgus group. No L-FKPext-valg was found in the varus group, as no L-FKPext-var were found in the valgus group. Nonetheless, a large variability in L-FKPext was detected in straight knees as previously depicted in Table 3a.

L-FKPflex expressed a low correlation with HKA in the MA workflow ($r = 0.29$), with a resulting statistical power of 0.87. L-FKP flex showed a larger lateral laxity for all groups (Table 3a). It was significantly different between varus and straight as well as between varus and valgus; however, it was not between straight and valgus. LPflex showed a uniform pattern with the majority expressing a L-FKPflex-lat in 98% of patients, only 1% of patients had L-FKPflex-neut, and 1% manifested L-FKPflex-med.

A low correlation ($r = 0.29$) was found between L-FKPext and L-FKPflex in MA, with a computed post hoc statistical power analysis of 0.87.

Comparison of L-FKP between B-FKP and MA

For L-FKPext significant differences were found between both workflows, for the total as well as for each subgroup. While the significance level was extremely high ($p < 0.000001$) for total ($d = 0.44$), varus ($d = 1.77$) and straight group ($d = 0.24$), it was only $p < 0.05$ for the valgus group ($d = 1.62$), most probably

due to the low sample size (13 patients in the valgus group). In all groups (73.3%), the gap difference was significantly reduced if patient-specific bony anatomy was applied compared with MA.

For L-FKPflex significant differences between B-FKP and MA workflow were again found for all groups (total: $d=0.96$; straight: $d=0.76$; varus: $d=1.71$), being least significant in valgus ($p<0.05$, $d=0.04$). This demonstrates that the difference in the tibia cut can cause a significant reduction of external rotation if the B-FKP workflow is applied and compared to MA. This is true if the femoral posterior condyle cut is planned parallel to the individual bony anatomy of the femoral posterior condyles.

DISCUSSION

The most important finding of this study is that L-FKP patterns and values are significantly affected by bony resections. Patient-specific bone resections (B-FKP workflow) are leading to minimal, non-significant gap differences in extension (L-FKPext) and flexion (L-FKPflex), minimising the need for soft-tissue release in the majority of cases. Larger variability of B-FKP workflow is shown in extension and flexion. MA workflow shows significant differences for L-FKPext and L-FKPflex within all CPAK subgroups, a high correlation with HKA in extension, and a very uniform lateral laxity pattern in 98% of cases in flexion. Both patterns lead to the need for soft-tissue release and increased external rotation in MA. The hypothesis that the personalised B-FKP workflow compared to MA shows significant differences for L-FKPext and L-FKPflex in all subgroups was confirmed.

In recent years, an increasing number of publications on bony phenotypes have been published, all showing a large variability of knees for the bony side [2, 5, 7, 11, 18, 25]. Based on these observations, patient-specific workflows, such as KA, rKA, iKA and FA, were reported [16, 17, 26, 27]. Reconstruction of bony phenotypes could therefore represent an important step towards individualised TKA. However, along with bony alignment, laxity patterns of the knees are also an important individual factor. Graichen et al. could recently show that a huge ligamentous variability in varus and valgus knees is common, and therefore not that one varus or valgus knee exists [3, 6]. So far, however, all studies are based on MA simulations with standardised 3° external rotation and 0° proximal tibia as well as distal femoral cut. Such standardised surgical concepts significantly influence the L-FKP of knees, therefore impacting joint line height, orientation, and stability throughout the range of motion. This is the first study, to simulate individual L-FKP patterns and values after implementing the patient-specific B-FKP. This included MPTA and LDFA for L-FKPext and MPTA

and individual rotation for parallel femoral posterior condyle cuts.

In general, it can be stated that after implementing B-FKP parameters, the need for soft tissue releases is reduced in the majority of knees. This is demonstrated by minimal and non-significant gap difference in extension, being most prominent for lower MPTA values. Overall, our simulations on real patient data show that in all CPAK groups, the need for release is none or minimal in almost 50% of knees. In severe varus deformities with an MPTA less than 85°, boundaries have reduced the potential of balancing the knee by bone cuts alone. These restricted boundaries were implemented in order to maintain the tibial tray within safety margins so that it would not impact implant survival. While incorporating individual bony phenotype parameters provides a more personalised approach, it still does not eliminate the challenge of soft tissue management in all cases. The observed variability in laxity patterns raises critical questions about whether all individual L-FKP patterns should be fully restored in every case. In particular, L-FKPext-valg and L-FKPflex-medlax patterns might be problematic, and therefore, adaptations of bone cuts might be considered in order to optimise the patient-specific L-FKP.

Changes of MPTA have a direct effect not only on L-FKPext, but also on L-FKPflex by altering the rotation of the femoral component when gap balancing workflows are considered. While a lower MPTA reduces the need for external rotation, an MPTA larger than 90° increases it. In contrast to MPTA, the posterior femoral cut, if parallel to the posterior condyles, is not different between workflows and therefore does not explain differences between B-FKP and MA. Therefore, the significant differences between L-FKPflex might be significantly influenced by MPTA differences. While in MA, 98% of knees showed L-FKPflex-latlax, this value was reduced to 56% in B-FKP workflows. Again, in B-FKP workflows, a larger variability of L-FKPflex patterns became obvious; 31% being L-FKPflex-neutr and 13% even manifesting medial laxity in flexion (L-FKPflex-medlax). Therefore, L-FKPflex-medlax seems to be problematic, as an increased internal rotation of the femoral component to counter-balance medial laxity might affect patella-femoral kinematics [22]. Making all gaps equal seems to be, based on present findings of laxity pattern variability, troublesome as well, given that almost 70% of knees are not L-FKPflex-neutr. This finding, however, has to be correlated with clinical outcome studies. In addition, the balancing goal of leaving the lateral flexion gap always larger than the medial, which is propagated by KA and rKA [13, 20, 23], might be correct in a higher percentage (56%) of knees, but still, 44% of patients seem to manifest a different laxity pattern in flexion. Determining femoral rotation based on individual L-FKPflex measurements rather than default external rotation

settings may offer a more tailored approach to knee balancing, while at the same time, it perfectly illustrates the type of surgical compromise that surgeons are currently facing.

While in MA the amount of deformity (HKA angle) correlates highly with the gap difference in extension [3, 19, 29], in B-FKP workflows this correlation is very low, signalling that bone cuts and alignment targets might be important confounding factors for post-operative patient satisfaction. In both workflows, a very low correlation between L-FKPext and L-FKPflex was found, showing that coronal bony anatomy (extension gap) does not correlate with flexion gap anatomy. Thus, individualised implants that depend on bone morphology can prove unpredictable when balanced gaps are aimed, as L-FKPflex patterns cannot be predicted based on bone morphology. Furthermore, there is a need to re-emphasise the problem of L-FKP-medlax again, as it might prove critical for post-operative patient satisfaction.

In this study, a knee-simulator (Knee-CAT) was used to analyse various bony and laxity parameters. As the aim of this study was to assess the effect of individualised bony reconstruction on L-FKP and compare this with MA, this simulator offered unique simulation options by allowing the inclusion of all relevant alignment and laxity parameters for such a post-operative analysis. A similar study can be performed with all modern robotic planning screens. However, such a study would need to be mainly performed intra-operatively, which would increase operating time and complexity. Further, almost all robotic systems are closed platforms, allowing only the assessment of the company's implementation. In Knee-CAT, all symmetric and asymmetric implants can be assessed, as long as geometrical dimensions are implemented.

Limitations of this study need to be mentioned. The fact, that posterior osteophyte removal in some knees might be incomplete at time of registration and simulation. Later osteophyte removal might affect gap data [28] and by that L-FKP values and L-FKP patterns would need to be adapted during surgery. Intra-operatively, pre-resection laxity might change following osteophyte, meniscus and cruciate ligament removal, thus requiring a second laxity assessment in order to avoid balancing errors. Another limitation might be the number of knees included in this study, however, the post-hoc statistical power analysis was performed in order to (1) inform the reader on the context of non-significant results, (2) provide a clearer understanding of the true effect size, (3) justify and validate the sample size for future studies. Nevertheless, further data acquisition is planned to improve the power of this L-FKP model. A final limitation might be the fact that this analysis was performed with the gap values in full (0°) extension. In this position, the tension of the posterior capsule might affect gap values.

CONCLUSION

Bony resections significantly impact L-FKPext and L-FKPflex. Personalised bony resections reduce the need for extensive soft tissue balancing in the majority of cases, particularly in extension, and minimise the need for excessive external rotation (L-FKPflex) of the femoral component compared to MA workflows. However, the observed variability in L-FKP patterns highlights the need for further refinement of individualised approaches. The patient-specific L-FKPext and L-FKPflex are important parameters which facilitate our understanding of individual balancing goals. Neither all gaps equal nor always creating lateral flexion gap laxity seems to be correct for all cases and might be modified individually. Future studies on larger sample sizes need to assess whether a correlation between L-FKP reconstruction and clinical outcomes exists.

AUTHOR CONTRIBUTIONS

Heiko Graichen: Conceptualisation; data curation; methodology; writing—original draft; writing—review and editing; resources. **George M. Avram:** Formal analysis; validation; writing—review and editing. **Francesco Zambianchi:** Visualisation; writing—review and editing. **Niklas M. Graichen:** Visualisation; writing—review and editing. **Fabio Catani:** Writing—review and editing; conceptualisation. **Sebastien Lustig:** Writing—review and editing; conceptualisation. **Michael T. Hirschmann:** Supervision; project administration; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The ethics statement is not available.

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